

Aniket Ghosh

ghosh.anik@northeastern.edu • Boston, MA • Portfolio • LinkedIn • GitHub

Education

M.S. in Artificial Intelligence, ML Concentration – Northeastern University – GPA: 4.0/4.0 Aug 2025 – May 2027
Courses: Foundations of AI, Algorithms, Actionable Interpretable Methods, Applied Programming for AI.

B.Tech in CS & Business Systems – Institute of Engineering & Management – GPA: 4.0/4.0 Jul 2020 – Jun 2024
Key Courses: Data Structures, Operating Systems, Database Management, Neural Networks, NLP

Publications & Patents

Intelligent Traffic Sign Detection Using YOLOv8 with Hybrid CNN Filtering *First Author*
Published – AISC 2024 • Springer “Information and Communication Technologies” series

Lightweight Hybrid DNN-GNN for Network Intrusion Detection with Adaptive Late Fusion *Co-author*
Published – IEEE • Patent Filed by SRM Institute of Science and Technology

Technical Skills

Languages: Python, C/C++, SQL, Bash

ML Frameworks: PyTorch, TensorFlow, scikit-learn, Hugging Face, OpenCV

GenAI & Agents: RAG, LangChain, CrewAI, OpenAI Agents SDK, MCP, ChromaDB, LoRA

MLOps: AWS, GCP, Docker, Kubernetes, Terraform, GitHub Actions, FastAPI, SLURM

Core Domains: Computer Vision, NLP, Multi-Agent Systems, Mechanistic Interpretability, Medical Imaging

Experience

AI/ML Engineer May 2026 – Present
Varosync, Inc., Boston, MA

- Building **hybrid ML pipelines** (classical + deep) and **knowledge-graph + embedding systems** for a drug-development AI platform, spanning representation learning, retrieval, KG reasoning, and **conditioning embeddings for generative models**, with reproducible runs and leave-one-group-out evaluation.
- Owning ML infrastructure across a hybrid cloud stack: **AWS, Nebius** (GPU training), and **Mithril** (inference/batch), with **Claude Code** and **Cursor** wired into the daily agentic loop.

AI Researcher Jun 2023 – May 2025
CMATER Lab, Jadavpur University, Kolkata

- Built an ML pipeline for histopathological cell segmentation. Achieved over 85% accuracy using color quantization and DBSCAN clustering, reducing false positives by 30% compared to a baseline Mask R-CNN model.

Computer Vision Research Intern Feb 2023 – Jul 2023
North-Eastern Hill University (Remote)

- Analyzed 50+ papers on sign language recognition to define quality and annotation standards for a new Indian Sign Language benchmark, drawing best practices from the Microsoft ASL and MNIST datasets.

Projects

Sufficient Cause Disambiguation (SCD) Interpretability Research • CS 7180 (PhD-level) • GitHub

- Proved probing accuracy and causal relevance are distinct properties on Llama-3-8B: separator tokens hit 1.000 LDA accuracy across all 32 layers yet drive 0% flip rate under causal patching.
- Built a mechanistic interpretability pipeline on **Llama-3-8B** (LDA + gradient attribution + causal patching); achieved **448× feature compression at 99.6% accuracy** and **100% prediction flip rate** at $\alpha=2$.
- Audited LLM resume screening: name-swapped resumes flip at **95.8%** and drive **62% of the model’s most uncertain Fit predictions** on otherwise-identical content.

Biomedical Knowledge Graph Link Prediction Healthcare AI • GitHub

- Developed a link prediction model on the BioRED biomedical corpus. Engineered graph features like PageRank to help a **Random Forest** model hit **0.94 ROC-AUC**, outperforming standard embeddings by **23%+**, and implemented **5-hop reasoning** to explain relationships.
- Benchmarked classical ML against **TransE knowledge graph embeddings** (0.61 ROC-AUC), showing engineered graph features substantially outperform learned embeddings on the smaller BioRED corpus.

Autonomous Multi-Agent Trading Simulation GitHub

- Built a multi-agent system with **4 autonomous AI traders**, **6 MCP servers**, and **44 distinct tools** using the OpenAI Agents SDK and integrated live market data via **Polygon.io** for autonomous portfolio management.

Achievements & Leadership

- **Teaching Co-Lead, AIDE Program:** Designing a **12-session ML lab** for philosophy PhD students at Northeastern’s AI and Data Ethics Summer Program (Jun–Aug 2026).
- **Graduate TA:** Lead PyTorch and deep learning labs for **40+ graduate students** in NLP at Northeastern University.
- **Academic Excellence:** Ranked **2nd out of 180+ students** in B.Tech; awarded the **Director’s Award**.
- **Volunteer Educator:** **7 years** teaching math & CS to **underprivileged students** at The Hope Foundation.
- **Graduate Student Senator:** Elected to Northeastern’s **Graduate Student Government**.